

Data Visualization

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Topics

1. Data Visualization
2. Data Visualization Golden Rules: Low Data/Ink Ratio and “No Chart Junk”
3. Tables
 1. Proper Data Sets
 2. Cross Tabulation
 3. PivotTables

Topics

1. Charts

1. Column and Bar Charts
2. Stacked Column or Bar Chart
3. Clustered column or Bar Chart
4. Line Charts
5. X-Y Scatter Charts
6. Bubble Charts

2. Conditional Formatting

3. Geographical Information System

4. Data Dashboards

Data Visualization

- Tables:
 - Summarize data
 - See exact numbers
 - Help interpret, analyze, and learn from the data
 - Used when numbers are in different units or magnitudes
 - Make precise comparisons
- Charts
 - Quick visual impression
 - See trends and patterns
 - Help interpret, analyze, and learn from the data
 - Make relative comparisons

Data Visualization

- Conditional Formatting
 - Some examples:
 - Mark numbers over a hurdle
 - Heat Maps
- Geographical Information System
- Dashboards
 - Important metrics presented visually and connected to an updateable data source

Effective Data Visualization

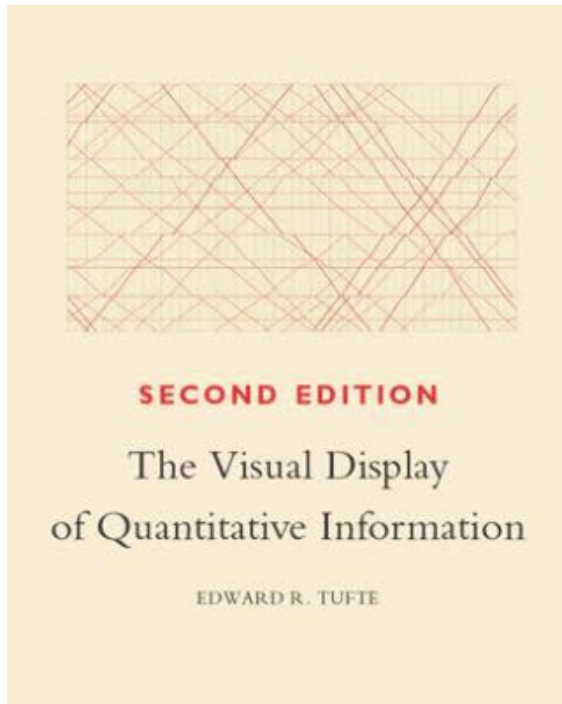
Allows You To Take Raw Data And:

- See patterns and trends
- Analyze data
- Convey your analysis to others
- Make your analysis easier for others to “see” and understand
- Learn from data
- Find errors

Visualization

- Research shows that humans can process visual images (like charts) faster than they can process rows of numbers
- Research shows that column and bar charts can convey differences more easily than pie charts

Edward R. Tufte



- Edward R. Tufte on visually portraying quantitative data:
 - Data-Ink Ratio should be high:
 - All ink in the chart or table should help deliver the message or the meaning of the data
 - Ink that serves no useful purpose must be removed
 - “No Chart Junk”
 - All elements in a chart should help deliver the message
 - In every chart, ask: does the element help? If the message is no, click the Delete key.

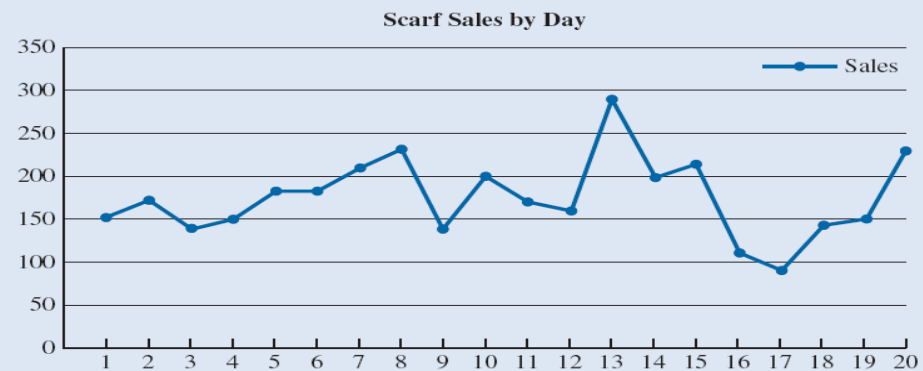
Data Visualization Golden Rule:

- Data Visualization Golden Rule:
 - **No extraneous elements in your table or chart**
- Implementing this rule means:
 - All elements in the chart or table should help deliver the message
 - “No extraneous elements” leads to more effective tables and charts
- Side note:
 - One of the hallmarks of great art is that great art has zero or near zero extraneous elements
- Benefit of following this rule:
 - Effective tables and charts allows you to better communicate your analysis to others

Low Data-Ink Ratio (Data/Ink)

From
Essentials of
Business
Analytics
textbook

Scarf Sales By Day			
Day	Sales	Day	Sales
1	150	11	170
2	170	12	160
3	140	13	290
4	150	14	200
5	180	15	210
6	180	16	110
7	210	17	90
8	230	18	140
9	140	19	150
10	200	20	230



Contains Chart
Junk

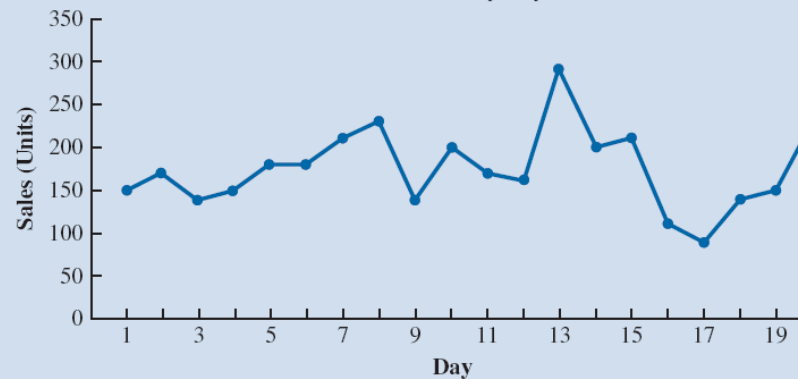
High Data-Ink Ratio (Data/Ink)

From
Essentials of
Business
Analytics
textbook

Scarf Sales By Day

Day	Sales	Day	Sales
1	150	11	170
2	170	12	160
3	140	13	290
4	150	14	200
5	180	15	210
6	180	16	110
7	210	17	90
8	230	18	140
9	140	19	150
10	200	20	230

Scarf Sales by Day



Does Not
Contain Chart
Junk

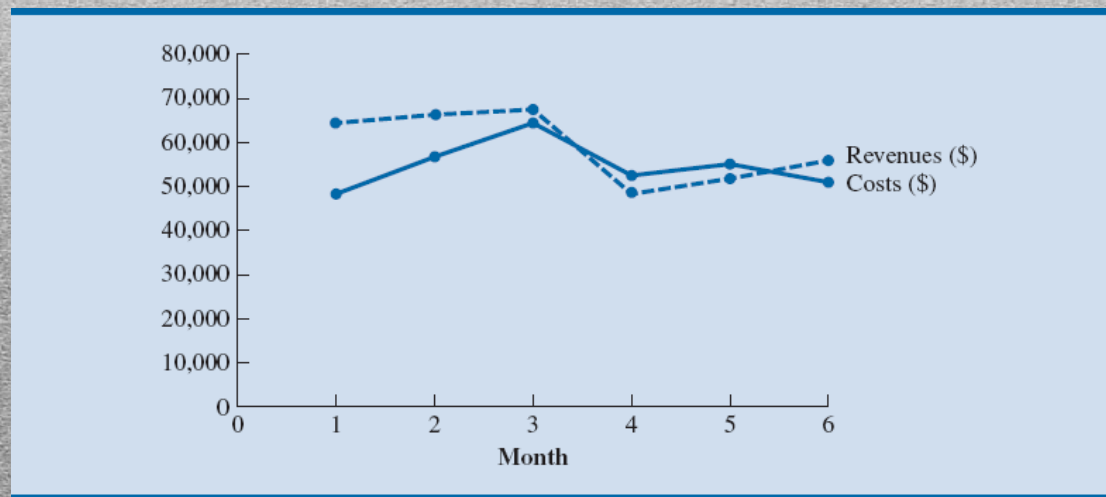
Exact Comparison Vs. Relative Comparison

From
Essentials of
Business
Analytics
textbook

Exact

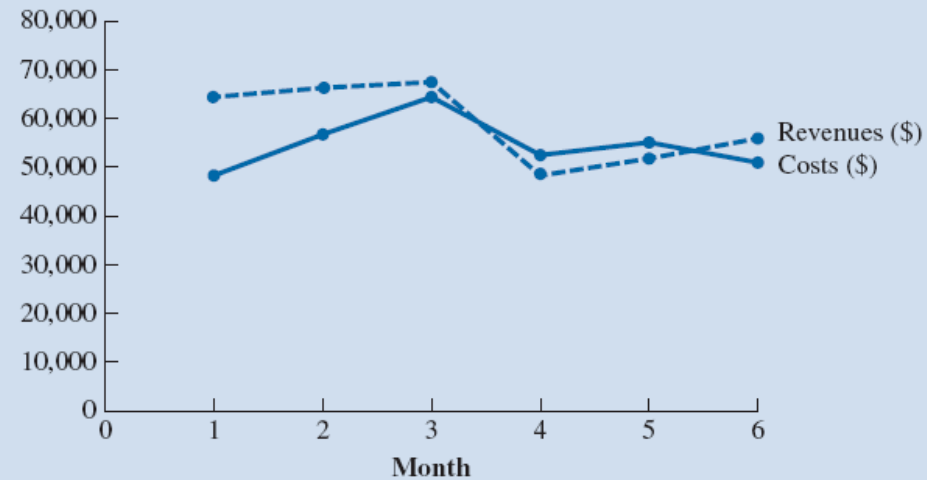
	Month						
	1	2	3	4	5	6	Total
Costs (\$)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues (\$)	64,124	66,128	67,125	48,178	51,785	55,687	353,027

Relative



Both: Exact Comparison & Relative

From
Essentials of
Business
Analytics
textbook



	Month						Total
	1	2	3	4	5	6	
Costs (\$)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues (\$)	64,124	66,128	67,125	48,178	51,785	55,687	353,027

Different Units: Use a Table

From
Essentials of
Business
Analytics
textbook

	Month						
	1	2	3	4	5	6	Total
Headcount	8	9	10	9	9	9	
Costs (\$)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues (\$)	64,124	66,128	67,125	48,178	51,785	55,687	353,027

Tables

Proper Data Sets

Cross Tabulation

PivotTables

Tables Design Principles

- Data-Ink Ratio should be high
- Textbook suggests:
 - Avoid using vertical lines in a table unless they are necessary for clarity.
 - Horizontal lines are generally necessary only for separating column titles from data values or when indicating that a calculation has taken place.
 - In large tables, light shading can be used to differentiate columns
 - Numbers should be right aligned (Right is the visual cue that it is a number)
 - Text should be left aligned (Left is the visual cue that it is a text)
 - All numbers should have same number of digits
 - Units must be indicated either with Number Formatting or Labels
 - Large numbers may be rounded to dollar or thousands or millions and so on

Table Design

- Author states that most people prefer Design D

Design A:

	Month						
	1	2	3	4	5	6	Total
Costs (\$)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues (\$)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits (\$)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design B:

	Month						
	1	2	3	4	5	6	Total
Costs (\$)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues (\$)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits (\$)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design C:

	Month						
	1	2	3	4	5	6	Total
Costs (\$)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues (\$)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits (\$)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design D:

	Month						
	1	2	3	4	5	6	Total
Costs (\$)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues (\$)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits (\$)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Different Shades Sometimes Helps A Viewer To Read Table

Revenues by Location (\$)	Month												Total
	1	2	3	4	5	6	7	8	9	10	11	12	
Temple	8,987	8,595	8,958	6,718	8,066	8,574	8,701	9,490	9,610	9,262	9,875	11,058	107,895
Killeen	8,212	9,143	8,714	6,869	8,150	8,891	8,766	9,193	9,603	10,374	10,456	10,982	109,353
Waco	11,603	12,063	11,173	9,622	8,912	9,553	11,943	12,947	12,925	14,050	14,300	13,877	142,967
Belton	7,671	7,617	7,896	6,899	7,877	6,621	7,765	7,720	7,824	7,938	7,943	7,047	90,819
Granger	7,642	7,744	7,836	5,833	6,002	6,728	7,848	7,717	7,646	7,620	7,728	8,013	88,357
Harker Heights	5,257	5,326	4,998	4,304	4,106	4,980	5,084	5,061	5,186	5,179	4,955	5,326	59,763
Gatesville	5,316	5,245	5,056	3,317	3,852	4,026	5,135	5,132	5,052	5,271	5,304	5,154	57,859
Lampasas	5,266	5,129	5,022	3,022	3,088	4,289	5,110	5,073	4,978	5,343	4,984	5,315	56,620
Academy	4,170	5,266	7,472	1,594	1,732	2,025	8,772	1,956	3,304	3,090	3,579	2,487	45,446
Total	64,124	66,128	67,125	48,178	51,785	55,687	69,125	64,288	66,128	68,128	69,125	69,258	759,079
Costs (\$)	48,123	56,458	64,125	52,158	54,718	50,985	57,898	62,050	65,215	61,819	67,828	69,558	710,935

Table Example 1 from Video:

Before

SalesRep	2015 Total Sales (\$)	2016 Total Sales (\$)	% Change	Customer Accounts	Years with Company
Maricela Merritt	30346.7	32291.9235	0.0641	8	1
Dick Fish	340821.2	317577.1942	-0.0682	88	11
Carl Levin	53363.33	42338.4660	-0.2066	18	2
Wilford Snell	366373.12	348750.5729	-0.0481	74	10
Mabelle Longo	264435.35	220539.0819	-0.1660	89	11
Tyrone Pham	483572.75	494527.0949	0.0227	113	15
Chin Smithe	56195.37	43135.5660	-0.2324	33	6
Terica Mcswain	247830.8	227136.9282	-0.0835	45	10
Gigi Wilke	296267	289749.1260	-0.0220	96	13
Brook Unger	228739.45	221991.6362	-0.0295	63	9

After

SalesRep	2015 Total Sales (\$)	2016 Total Sales (\$)	% Change	Customer Accounts	Years with Company
Maricela Merritt	30,347	32,292	6.41	8	1
Dick Fish	340,821	317,577	-6.82	88	11
Carl Levin	53,363	42,338	-20.66	18	2
Wilford Snell	366,373	348,751	-4.81	74	10
Mabelle Longo	264,435	220,539	-16.6	89	11
Tyrone Pham	483,573	494,527	2	113	15
Chin Smithe	56,195	43,136	-23.24	33	6
Terica Mcswain	247,831	227,137	-8.35	45	10
Gigi Wilke	296,267	289,749	-2.2	96	13
Brook Unger	228,739	221,992	-2.95	63	9

Table Example 2 from Video:

GDP/breakdown at constant 2005 prices in US Dollars (all countries)					
Country	2000 (\$M)	2001 (\$M)	2002 (\$M)	2003 (\$M)	2004 (\$M)
Argentina	201,362,716,310.6	193,058,437,457.6	172,025,773,378.4	187,227,757,461.7	204,134,379,753
Belgium	355,035,280,793.6	358,316,763,281.8	363,904,793,271.8	367,143,188,125.1	379,753,855,028
Brazil	768,855,134,604.8	778,351,115,397.1	799,656,363,540.0	808,825,387,988.6	855,028,128,141
Canada	1,030,602,342,820.8	1,045,003,504,340.3	1,073,203,622,405.6	1,094,586,353,832.4	1,128,141,198,302
Finland	173,902,129,035.5	184,544,368,412.7	187,645,398,103.7	191,387,603,840.2	198,302,166,756
France	2,029,383,087,101.4	2,063,666,218,372.3	2,092,809,530,315.8	2,109,358,712,630.0	2,166,756,2,837,510
Germany	2,777,363,467,542.2	2,824,447,709,015.7	2,824,751,452,450.8	2,804,398,873,608.1	2,837,510,245,479
Greece	204,352,392,090.7	212,611,323,593.5	219,335,320,011.9	233,894,816,694.0	233,895,152,612
Greenland	1,527,406,281.9	1,546,783,610.3	1,531,767,530.8	1,525,352,532.2	1,526,153,151
Ireland	165,331,285,098.8	174,682,021,353.8	184,880,332,774.0	190,346,718,090.3	190,347,199,062
Italy	1,768,788,613,639.8	1,800,146,122,463.3	1,804,660,622,792.6	1,807,414,564,390.3	1,807,415,1,836,032
Luxembourg	31,791,026,215.0	32,426,876,585.6	33,483,155,847.4	33,887,728,193.8	33,888,35,557
Mexico	788,246,519,332.4	787,387,763,283.2	794,070,620,626.4	805,110,296,054.6	805,110,839,005
Norway	276,834,371,648.0	282,668,537,410.8	286,732,484,021.1	283,363,967,486.2	289,370,300,826
Panama	12,523,862,775.4	12,595,783,328.5	12,876,523,746.5	13,418,086,116.4	13,418,14,427
Peru	61,705,803,374.4	62,087,078,737.9	65,473,015,607.0	68,193,381,855.3	68,200,71,581
Poland	262,392,731,637.2	266,162,582,612.6	270,004,636,092.5	279,623,662,616.1	279,624,293,984
Portugal	188,374,536,610.5	192,646,303,601.8	194,127,312,503.9	192,314,402,139.0	192,314,195,798
Saudi Arabia	258,610,834,969.3	260,026,371,690.8	260,359,146,865.2	280,300,688,978.8	280,301,306,240
Slovakia	38,278,398,188.2	39,550,521,367.3	41,415,097,698.9	43,656,221,714.6	43,656,45,944
Slovenia	30,461,740,243.5	31,360,185,920.5	32,563,216,768.6	33,488,736,683.0	33,489,34,946
Spain	379,526,062,385.8	1,018,717,818,112.4	1,048,054,809,786.3	1,081,462,185,898.5	1,081,462,1,115,709
Sweden	341,717,335,402.3	347,059,840,287.9	354,256,394,050.4	362,708,031,380.5	362,708,378,379
Switzerland	378,377,143,051.9	383,850,698,282.4	384,401,146,039.9	384,592,234,121.6	384,592,395,526
Turkey	386,584,134,135.9	364,558,594,063.9	387,023,401,185.4	407,407,526,172.1	407,408,445,552
United Arab Emirates	139,150,824,019.6	141,065,061,316.3	144,497,703,072.0	157,214,336,283.5	157,214,172,254
United Kingdom	2,087,472,757,637.1	2,143,094,196,538.3	2,195,648,906,230.7	2,290,071,474,844.1	2,290,071,2,346,267
United States	11,553,315,901,319.6	11,666,074,165,730.0	11,874,444,226,541.1	12,207,733,297,850.8	12,207,733,12,663,888
Venezuela	128,278,574,764.1	132,632,652,467.3	120,887,172,490.0	111,512,003,542.0	111,512,131,904

GDP/breakdown at constant 2005 prices in US Dollars (all countries)					
Country	2000 (\$M)	2001 (\$M)	2002 (\$M)	2003 (\$M)	2004 (\$M)
Argentina	201,363	193,058	172,026	187,228	204,134
Belgium	355,035	358,317	363,905	367,143	379,753
Brazil	768,855	778,351	799,656	808,825	855,028
Canada	1,030,602	1,045,010	1,073,204	1,094,586	1,128,141
Finland	173,902	184,545	187,646	191,388	198,302
France	2,029,383	2,063,666	2,092,810	2,109,359	2,166,756
Germany	2,777,363	2,824,448	2,824,751	2,804,399	2,837,510
Greece	204,353	212,612	219,336	233,895	245,479
Greenland	1,527	1,547	1,532	1,526	1,531
Ireland	165,331	174,682	184,880	190,347	199,062
Italy	1,768,789	1,800,146	1,804,661	1,807,415	1,836,032
Luxembourg	31,791	32,427	33,489	33,888	35,557
Mexico	788,247	787,388	794,071	805,110	839,005
Norway	276,834	282,669	286,732	289,370	300,826
Panama	12,524	12,596	12,877	13,418	14,427
Peru	61,706	62,087	65,473	68,200	71,581
Poland	262,393	266,163	270,005	279,624	293,984
Portugal	188,375	192,647	194,128	192,314	195,798
Saudi Arabia	258,611	260,027	260,359	280,301	306,240
Slovakia	38,278	39,551	41,415	43,656	45,944
Slovenia	30,462	31,360	32,563	33,489	34,946
Spain	379,526	1,018,718	1,048,055	1,081,462	1,115,709
Sweden	341,717	347,060	354,256	362,708	378,379
Switzerland	378,377	383,851	384,401	384,592	395,526
Turkey	386,584	364,553	387,023	407,408	445,552
United Arab Emirates	139,151	141,065	144,498	157,214	172,254
United Kingdom	2,087,473	2,143,094	2,195,649	2,290,071	2,346,267
United States	11,553,316	11,666,074	11,874,444	12,207,733	12,663,888
Venezuela	128,279	132,633	120,887	111,512	131,904

Cross Tabulation

- “Calculations with two or more criteria or conditions”
 - Allows you to compare two or more variables and shows relationship between variables (can be categorical or quantitative)
- PivotTables are perfect for creating Crosstabulations
 - Variable/Field/Criteria/Condition as Row Header
 - Variable/Field/Criteria/Condition as Column Header
 - Intersection is a calculation done with two or more criteria/conditions
 - Sum
 - Count
 - "Show Values as" and then choose "% of Grand Total", "% of Row Total", "% of Column Total"
 - PivotTables and the PivotTable grouping feature were covered in chapter 02

Chart Types

- Column and Bar Charts
- Stacked Column or Bar Chart
- Clustered column or Bar Chart
- Line Charts
- X-Y Scatter Charts
- Bubble Charts
- Heat Map
- Graphical Information Systems
- Textbook & Authors finally reflect the truth:
 - Pie charts are not as effective as column/bar charts
 - 3-D charts (not bubble charts) are “Chart Junk”

What do Charts do?

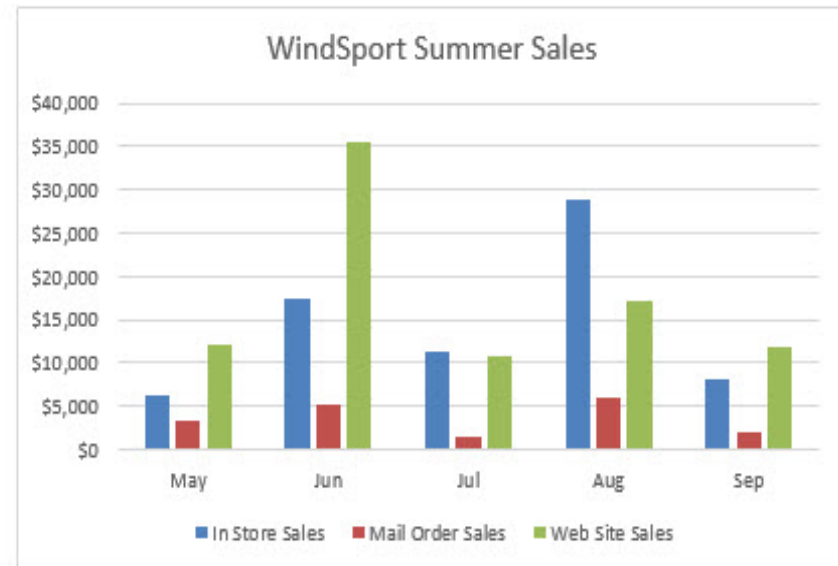
- Visually portray Quantitative data (number data).
- Give a quick impression of the number data.
- Create a picture that can communicate more quickly than just the numbers alone.
- Charts allow you to see patterns or trends that you may not be able to see if you are looking at just the number data.
- Allows you to make relative comparisons more quickly than if you are using a table

Effective charts:

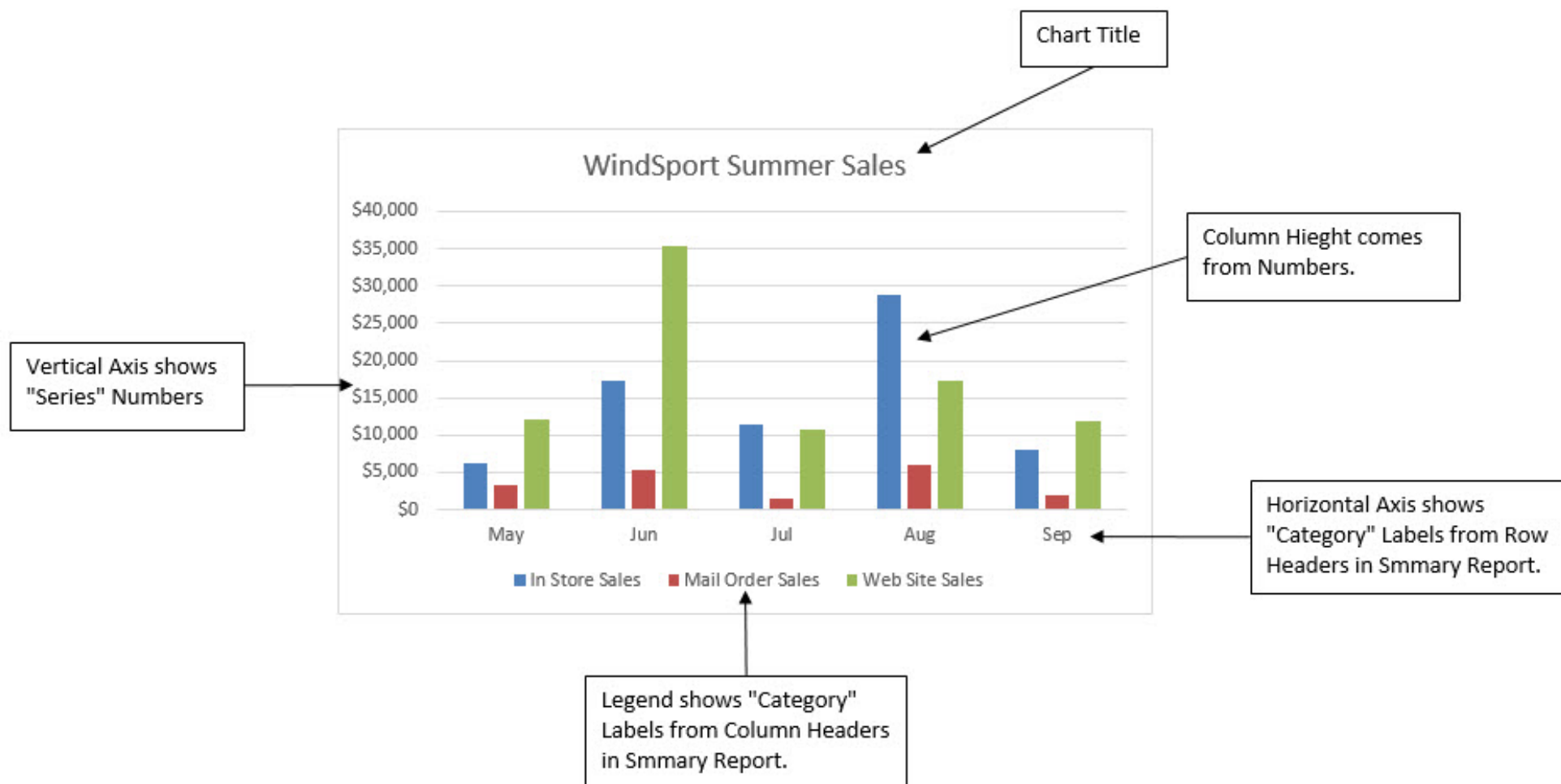
- Number data AND labels for the number data
- No “Chart Junk”
 - Chart Junk:
 - Unnecessary Repetition
 - Chart elements that do not contribute to the message
 - Chart elements that make the chart look busy
 - Too many different colors
 - Patterns that are distracting
 - 3-D effects that are not necessary or misleading

Charts Usually Come From Summarized Data Tables

WindSport Summer Sales				
Month	In Store Sales	Mail Order Sales	Web Site Sales	Total
May	\$6,206	\$3,275	\$12,016	\$21,497
Jun	\$17,351	\$5,328	\$35,371	\$58,050
Jul	\$11,360	\$1,555	\$10,822	\$23,737
Aug	\$28,722	\$5,913	\$17,243	\$51,878
Sep	\$7,995	\$1,913	\$11,764	\$21,672
Total	\$71,634	\$17,984	\$87,216	\$176,834



Excel Chart Terminology



Column and Bar Charts

- Show differences (in numbers) across categories (labels)
- Height of columns convey number
- Good for Categorical Data:
 - Count (Frequency) by category
 - Used to show Frequency Distribution or Relative/Percent Frequency Distribution for Categorical Data
 - For categorical data, order of categories conveys no info
 - There are "gaps" between columns to indicate that the data is categorical or a discrete quantitative variable. Columns do not touch.
- Research shows that Column or Bar Charts convey relative differences more effectively than Pie Charts
- In recent years data analysis practitioners tend to use Column or Bar Charts rather than Pie Charts

Stacked Column or Bar Chart

- Good for displaying crosstabulation.
- Great chart for cross tabulated data from PivotTable
- **Emphasis is on comparing the categories listed in the horizontal axis**
- Excel:
 - If the number of row headers are equal or greater than to the number of column headers, row headers show up on horizontal axis and column headers in legend. If not, they are reversed. (You can switch this with the Switch button in the Chart Tools Design Ribbon Tab)

Clustered Column or Bar Chart

- Good for displaying crosstabulation.
- Great chart for cross tabulated data from PivotTable
- **Emphasis is on comparing the categories listed in the legend**
- Excel:
 - If the number of row headers are equal or greater than to the number of column headers, row headers show up on horizontal axis and column headers in legend. If not, they are reversed. (You can switch this with the Switch button in the Chart Tools Design Ribbon Tab)

Multiple Charts

- An alternative to Stacked or Clustered Column/Bar Charts that breaks the variables in the legend into separate charts
- Important to have consistent axes units and/or categories
- May make the data less cluttered
- Takes more space
- When you have many variables or categories, multiple charts may be cleaner and easier to read
- Both column/bar and scatter charts can be effectively used for multiple charts
- Scatter charts matrix
 - Useful for analyzing relationships among variables

Line Charts

- One number on vertical axis, category on horizontal axis
- Great for show trends over time

X-Y Scatter Charts

- Chart that shows the relationship between two quantitative variables
- One number on vertical axis, one number on horizontal axis
 - Horizontal Axis = Independent Variable = x .
 - Vertical Axis = Dependent Variable = $f(x) = y$
- Always put X values in Left Most Column in the Table of Data
- Add Regression Line and Equation and R Square:
 - Right-click plotted scatter markers
 - Add Trendline
 - Select Linear
 - Check check box for Show Equation
 - Check check box for R Square
- Overcome a common mistake by Excel users:
 - Use X-Y Scatter Plot Chart, not Line Chart when plotting X-Y Scatter Data

Bubble Charts (3-D Graph)

- Method of visualizing 3 variables in a 2-dimensional chart
- First column of table = horizontal axis
- Second column in table = vertical axis
- Third column = size of bubble (marker of the intersection of the horizontal and vertical axis numbers).

Conditional Formatting

- Examples:
 - Highlight value over a hurdle
 - Heat Map
 - Conditional Formatting to visualize data into thirds or quartiles by using a certain color for each section of the sorted data.
 - To create a Heat Map use: Conditional Formatting, “Color Scales”
 - Examples given in chapter 2 in the video at YouTube:
 - [Basic Excel Business Analytics #14: Logical Formulas & Conditional Formatting to Visualizing Data](#)

Geographical Information System

- Maps that present data and statistics
- Excel 2016: Insert Ribbon Tab, 3-D Maps

Data Dashboard

- Data Dashboard
 - Data Visualization Tool that illustrates multiple metrics or KPIs (Key Performance Indicators) and automatically updates as new data becomes available
- Effective Dashboards:
 - Presents timely summary data or metrics/KPIs
 - Metrics/KPIs should be useful for the user/decision maker
 - Dashboard should inform rather than overwhelm
 - Should call attention to unusual metrics/KPIs that require attention or are of interest

Histograms

- Used to show frequency distribution of continuous quantitative data over a set of class intervals (lower and upper limit for each category)
- Column or Bar Charts where columns are touching to indicate that the variable is continuous
- Columns touch to indicate that no numbers can fit between classes. "No numbers can fit between columns - no gaps"
- Height of columns convey count
- Order of classes is important to help reveal shape of data, or distribution of data
- Skew of Histograms
 - What does the distribution of Histogram Columns look like?
 - Skew Left or Negative means a few short Histogram Columns are on the low end (pull mean down)
 - Skew Right or Positive means a few short Histogram Columns are on the high end (pull mean up)
 - No Skew means the distribution is bell shaped or nearly bell shaped

Charts to show Distribution of Data:

- Column/Bar/Pie: Used to show Frequency Distribution or Relative/Percent Frequency Distribution for Categorical Data or Discrete Quantitative Data
- Dot Plot: Used to show distribution of quantitative data over entire range of data
- Histogram: Used to show frequency distribution of continuous quantitative data over a set of class intervals (lower and upper limit categories)
- Stem and Leaf: Allows you to see the shape of distribution for a quantitative variable without losing the identity of each value

Charts to show comparisons:

- Clustered Column: Emphasis is on comparing the categories listed in the legend
- Stacked Column: Emphasis is on comparing the categories listed in the horizontal axis
- 100% Stacked Column: Each column is Similar to a pie chart. Emphasis is on comparing percentages for categories in Legend